

The Health Research Lifecycle

Level 1 - Research Foundations | Student Learning Pack

How to use this pack

Read the teaching sections first, complete the worked example, do the short practice task, then take the basic knowledge check in RSRE. External references are optional further learning; you should be able to answer the core assessment from the Academy material.

Lesson 1: See the whole journey

The health-research lifecycle is a connected sequence rather than a single event. A typical journey is: identify a problem → review existing evidence → define the question → choose a design → plan ethics and governance → collect data → manage and analyze data → interpret findings → communicate or publish → preserve the record and learn from the work.

The stages influence one another. A poorly defined question creates problems for the design. A weak data plan creates problems for analysis. Poor reporting makes useful findings difficult to trust or reproduce.

Lesson 2: From question to plan

Take a simple question such as “What factors are associated with missed antenatal-care visits in a district?” Map the next five steps. Start with the evidence already available, then identify the study population, possible variables, ethical considerations and the kind of result you need. The goal is not to choose a perfect method yet; the goal is to see how each decision connects to the next.

Lesson 3: Governance is part of the lifecycle

Research involving people, identifiable data, biological materials or interventions may require ethics review, institutional permission, regulatory authorization or other governance processes. These should be planned before activities that require approval begin.

A responsible researcher also plans data security, authorship, documentation, conflicts of interest and dissemination early rather than at the end.

Lesson 4: Worked example

A student notices low vaccination knowledge among peers. First, check whether the question has already been answered. Next, define exactly what “knowledge” means. Then choose a suitable study design, think about recruitment and consent, plan the questionnaire, decide how data will be stored, and only then conduct the study. After analysis, communicate both strengths and limitations.

Write the lifecycle for your own example in ten short steps.

Lesson 5: Ready for assessment

You should now recognize research as a process with linked decisions. You do not need to memorize one rigid sequence; different studies may revisit earlier stages. The important skill is understanding why each stage exists and how evidence, ethics, methods and communication connect.

Competency check

After completing this module, you should be able to explain the core idea in your own words, identify a correct and incorrect application, and make a basic research decision using the concept.

RSRE note: The Academy is optional. Experienced researchers may use the diagnostic/competency pathway and enter other RSRE pillars directly.